

EBMF Diagnostic: Does Survival Signal Exist in Merged TCGA + CPTAC?

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1 Motivation

Training SSBMF on the merged TCGA_PAAD (RNA-seq) + CPTAC (proteomics) cohort consistently produces all factor coefficients $\hat{\beta}_k = 0$: no factor is selected as prognostic under the Bayesian survival objective. Before debugging the model, a more basic question must be answered:

Is the absence of signal a data problem or a model problem?

Two explanations are plausible:

1. *Data problem.* The RNA-seq vs. proteomics platform difference dominates all variance in the merged matrix; batch structure consumes the gene-set, leaving no detectable survival signal regardless of the analysis method.
2. *Model problem.* Survival signal exists in the data, but SSBMF's objective weighting, prior aggressiveness, or CAVI convergence behavior prevents it from reaching the β coefficients.

This diagnostic distinguishes between the two by running **purely unsupervised factor analysis** — Empirical Bayes Matrix Factorization (EBMF via `flashier`) and PCA — on the same merged training matrix used for SSBMF, then asking whether any resulting factor is univariately associated with survival via a Cox proportional-hazards model.

2 Methods

Data. v2-preprocessed merged training matrix: TCGA_PAAD ($n = 144$) + CPTAC ($n = 129$), $p = 2,000$ genes. Preprocessing: intersect raw gene universes (17,900 genes), $\log_2(x+1)$ transform (RNA-seq only), quantile normalization across all samples jointly, top-2,000 by merged-matrix variance, per-subject rank transform.

EBMF. `flashier::flash()` with per-gene residual variance (`var_type = 2`) and a greedy add + backfit strategy (`greedy_Kmax = 20`). Loadings extracted via LDF decomposition with unit- L_2 -norm columns so all factors are on a comparable scale.

PCA. `prcomp(Y, rank. = 20, center = TRUE)`.

Survival association. For each factor k , a univariate Cox model $h(t) = h_0(t) \exp(\ell_k \gamma_k)$ is fit where ℓ_k is the n -vector of subject loadings. Reported: log-hazard ratio $\hat{\gamma}_k$, hazard ratio (HR), Wald z -statistic, p -value, and concordance (C-index). Significance threshold: $p < 0.05$.

3 Results

3.1 EBMF

EBMF converged on **20 factors**. Of these, **5 are univariately associated with overall survival** at $p < 0.05$ (Table 1). The strongest association is EBMF6 (C-index = 0.63, $p = 3 \times 10^{-6}$), followed by EBMF4 and EBMF16. EBMF1 has an enormous point estimate ($\text{HR} \approx 4 \times 10^8$) with $p = 0.35$ — a signature of a near-collinear batch factor (very large loadings concentrated in one cohort, unstable Cox fit).

Table 1: Univariate Cox results for each EBMF factor ($K = 20$). Factors with $p < 0.05$ are marked in the final column.

Factor	log-HR	HR	z	p-value	C-index	p < 0.05
EBMF1	33.637	4.058e+14	0.935	0.35	0.536	
EBMF2	0.133	1.142	0.093	0.926	0.526	
EBMF3	3.228	25.22	2.160	0.0308	0.572	
EBMF4	-4.446	0.0117	-2.780	0.00544	0.578	
EBMF5	-1.617	0.1984	-1.082	0.279	0.550	
EBMF6	6.333	562.8	4.653	3e-06	0.628	
EBMF7	0.643	1.902	0.463	0.643	0.507	
EBMF8	-0.805	0.447	-0.585	0.559	0.487	
EBMF9	1.353	3.871	0.928	0.353	0.538	
EBMF10	-0.258	0.7728	-0.188	0.85	0.517	
EBMF11	0.658	1.931	0.421	0.674	0.517	
EBMF12	-0.582	0.5591	-0.385	0.7	0.515	
EBMF13	-2.675	0.0689	-1.730	0.0837	0.544	
EBMF14	-1.414	0.2433	-1.253	0.21	0.521	
EBMF15	0.917	2.503	0.660	0.509	0.515	
EBMF16	3.994	54.24	2.764	0.00571	0.570	
EBMF17	-3.246	0.0389	-2.240	0.0251	0.551	
EBMF18	-0.920	0.3987	-0.643	0.52	0.521	
EBMF19	0.931	2.536	0.674	0.5	0.493	
EBMF20	-0.060	0.9417	-0.042	0.966	0.516	

The cohort-stratified loading heatmap (Figure 1) shows that EBMF separates samples along both a strong cohort-collinear axis (consistent with a platform batch factor) and several axes that mix TCGA and CPTAC samples — the latter being the factors most likely to carry biological rather than technical signal.

Figure 2 annotates the same heatmap with Cox significance: the column strip shows $-\log_{10}(p)$ for each factor, with the active/significant threshold at $-\log_{10}(0.05) \approx 1.3$.

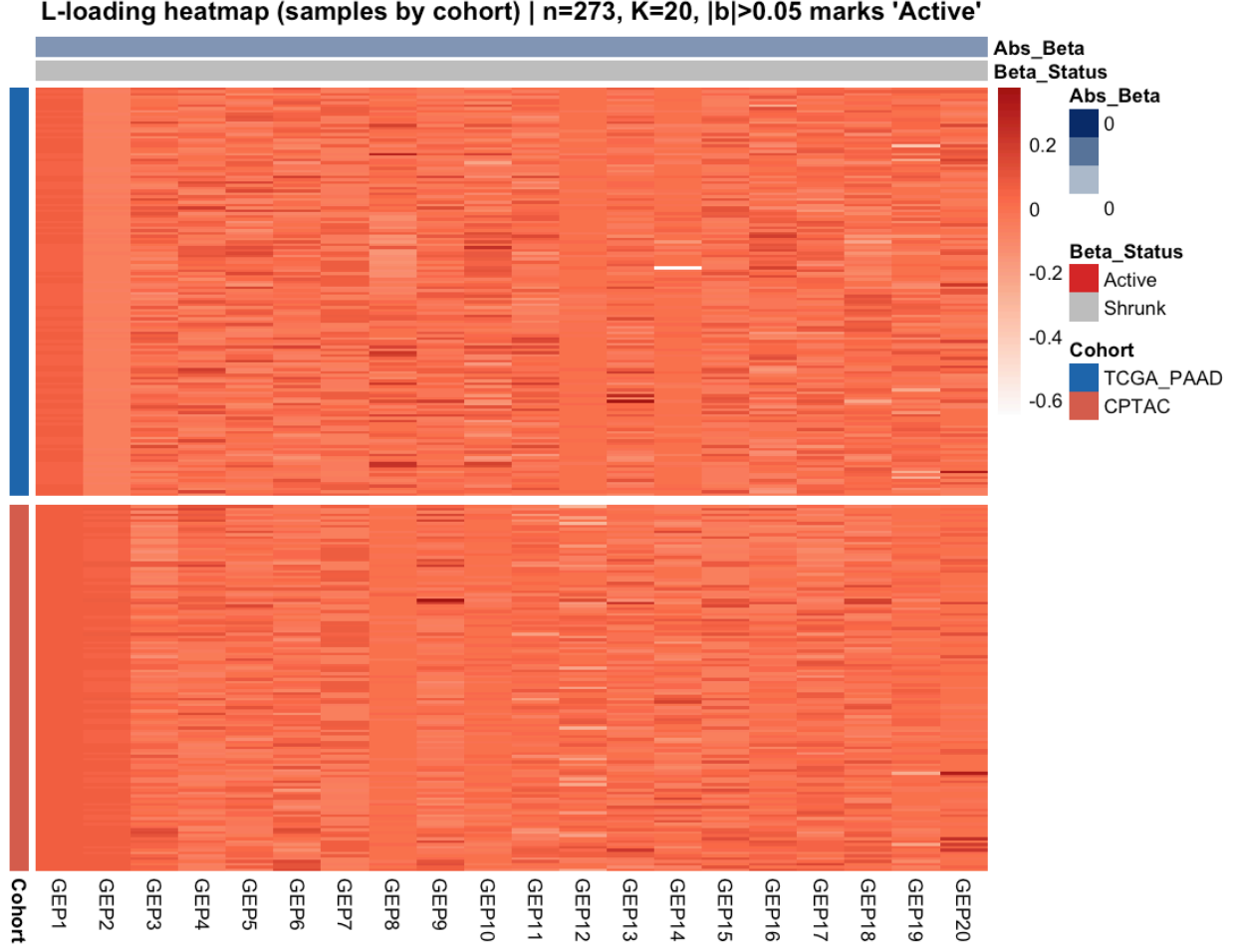


Figure 1: EBMF loading heatmap. Rows = subjects (sorted by cohort), columns = factors GEP1–GEP20. Row annotation (left) = cohort membership. No column annotation is shown because EBMF is unsupervised (is not estimated). Factors with the strongest within-cohort structure are likely batch-driven; factors that mix cohorts are candidates for biologically meaningful signal.

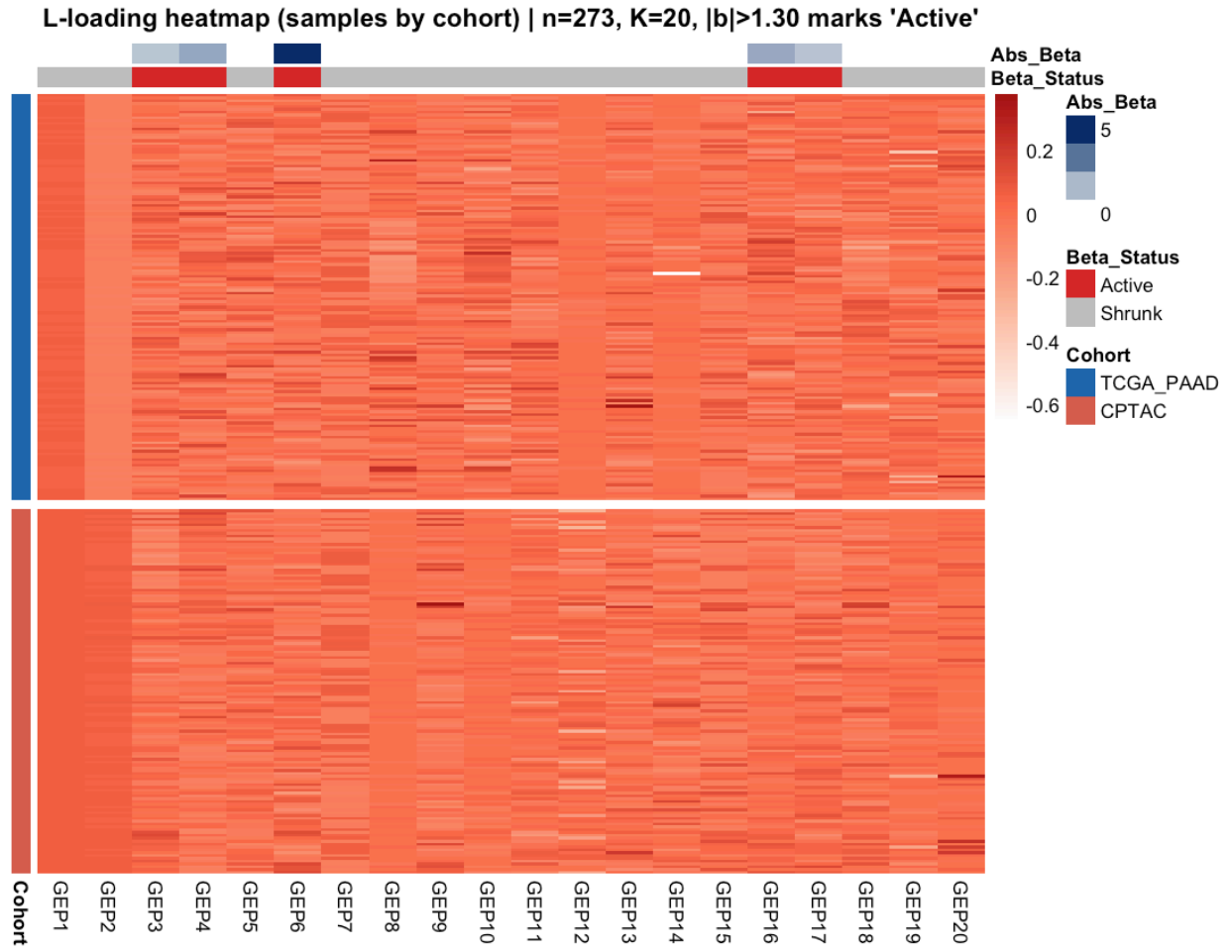


Figure 2: EBMF loading heatmap with Cox-significance annotation. Column annotation strip: orange–blue gradient = $-\log_{10}(\text{p-value})$ from univariate Cox model; ‘Active’ (red) = $p < 0.05$, ‘Shrunk’ (gray) = $p \geq 0.05$.

3.2 PCA

PCA finds **4 of 20 components** significantly associated with survival (Table 2). PC5 is the strongest (C-index = 0.61, $p = 1.5 \times 10^{-4}$). The concordance between EBMF and PCA (5/20 vs. 4/20, overlapping in significance rank) reinforces that the signal is real rather than an artifact of the EBMF model structure.

Table 2: Univariate Cox results for the top 20 PCA components.

Factor	log-HR	HR	z	p-value	C-index	p < 0.05
PC1	0	1	0.107	0.915	0.529	
PC2	0	1	1.905	0.0567	0.565	
PC3	0	0.9999	-2.768	0.00563	0.585	
PC4	0	1	-1.438	0.15	0.538	
PC5	0	1	3.797	0.000147	0.615	
PC6	0	1	-0.150	0.88	0.537	
PC7	0	1	-0.564	0.573	0.537	
PC8	0	1	1.972	0.0486	0.564	
PC9	0	1	-0.663	0.507	0.519	
PC10	0	1	0.959	0.337	0.524	
PC11	0	1	0.156	0.876	0.501	
PC12	0	1	1.882	0.0599	0.543	
PC13	0	1	0.698	0.485	0.534	
PC14	0	0.9999	-1.777	0.0756	0.515	
PC15	0	0.9999	-1.397	0.162	0.541	
PC16	0	1	0.331	0.741	0.517	
PC17	0	1	0.939	0.348	0.525	
PC18	0	1	0.375	0.707	0.494	
PC19	0	0.9998	-2.274	0.023	0.552	
PC20	0	0.9999	-1.007	0.314	0.512	

4 Takeaways

The central question is answered: **survival signal exists in the merged v2 data**. Five EBMF factors and four PCA components are univariately associated with overall survival at $p < 0.05$, with the strongest reaching C-index ≈ 0.63 using a single factor’s loadings alone. The failure of SSBMF to assign non-zero $\hat{\beta}$ to any factor is therefore a **model problem**, not a data problem.

Two follow-on observations sharpen the diagnosis:

1. **Lambda increase collapses the model.** A concurrent lambda sweep ($\lambda \in \{1, 5, 10, 20\}$) found that $\lambda = 5$ (and likely higher values) causes the CAVI to degenerate: $\hat{L} \rightarrow 0$ as well as $\hat{\beta} \rightarrow 0$. Naively amplifying the survival objective overshoots and destabilizes the factor-loading updates rather than pulling $\hat{\beta}$ away from zero.
2. **A strong batch factor coexists with survival-associated factors.** EBMF1 has loadings concentrated in one cohort and an unstable Cox estimate — the expected signature of an RNA-seq vs. proteomics platform factor. The fact that *other* factors (EBMF3, 4, 6, 16, 17) simultaneously associate with survival suggests the batch and biological signals occupy distinct

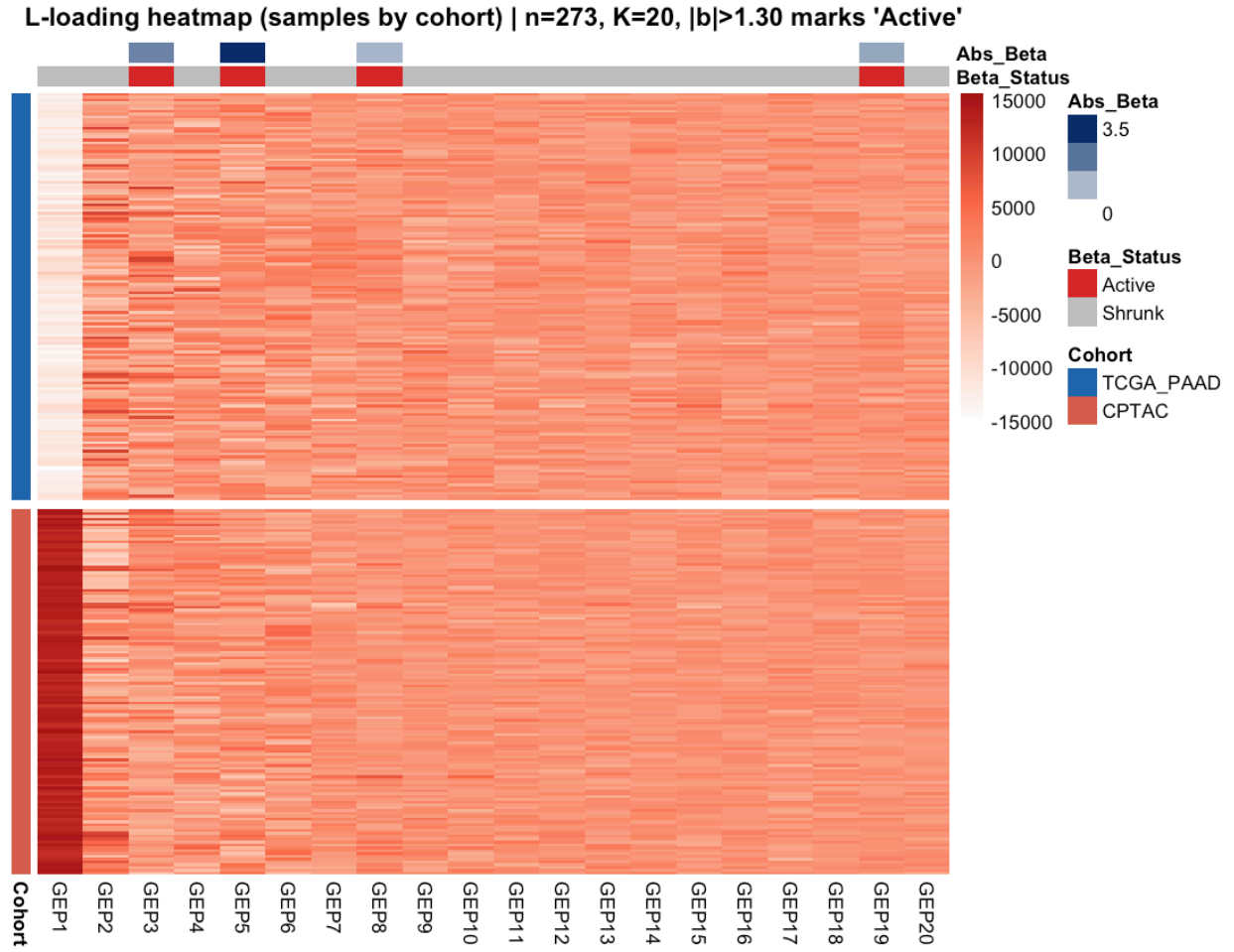


Figure 3: PCA loading heatmap with Cox-significance annotation. Same layout as Figure 2. Column strip = $-\log_{10}(\text{p-value})$; 'Active' = $p < 0.05$.

directions in the gene space. The challenge for SSBMF is to correctly weight the biological directions under the survival objective without being swamped by the larger-variance batch direction.

Recommended next steps:

- Investigate the CAVI update for $\mathbf{L}$ to understand why the survival gradient pushes $\hat{L}$ to zero rather than refining it toward the biologically informative factors identified by EBMF.
- Consider **EBMF-initialized SSBMF**: warm-start $\hat{L}$ and $\hat{F}$ from the EBMF solution and only optimize $\hat{\beta}$ from that fixed point. This sidesteps the initialization problem and directly tests whether the β update is capable of selecting the EBMF-identified survival factors.
- Compare factor loadings between EBMF6/EBMF4 (strongest survival association) and the SSBMF GEPs from single-cohort training to understand whether these are the same biological programs or orthogonal signals.